

Engineering Documentation

nat-os

DOCUMENT	REVISION	STATUS	CLASSIFICATION
nat-os	—	—	Internal

Hardware: developed and verified on the ESP32-2432S028R ("Cheap Yellow Display")

Renamed from `cyd-os` . Document numbering and section numbering are unchanged — [UM-CYDOS-014 §5.2](#) in an older commit message is [UM-NATOS-014 §5.2](#) here. The rename reflects that only the pin maps and the panel and touch drivers are specific to the Cheap Yellow Display; everything above them assumes an ESP32 and nothing more.

Purpose of this set

These reports record the design of nat-os in enough detail that the project can be picked up cold — by someone else, or by the author after six months away — without re-deriving decisions from the source.

Each report states **what was decided, what it was measured against, and what remains unverified**. Claims that were tested on hardware are marked as such. Claims taken from documentation or reasoning are marked separately, because on a from-scratch kernel the difference between "this is true" and "this should be true" is the difference between a working boot and a silent reboot.

Index

ID	Title	Covers
UM-NATOS-001	System Architecture and Scope	Layer model, what is built vs borrowed, the isolation problem, why a bytecode VM
UM-NATOS-002	Boot Chain and Image Format	ROM → 2nd stage → kernel, image header, measured boot trace
UM-NATOS-003	Xtensa ABI Selection	Windowed vs call0, codegen evidence, consequences for the scheduler
UM-NATOS-004	Memory Map and Allocation Policy	ESP32 regions, our layout, the bootloader overlap risk
UM-NATOS-005	Build and Flash Pipeline	Toolchain, flags and why each one, reproducing a build, why not PlatformIO
UM-NATOS-006	Milestone 0 Verification Report	Test method, captured output, pass/fail per assertion
UM-NATOS-007	Development Roadmap M1–M5	Each milestone, its risks, and its exit criteria
UM-NATOS-008	Milestone 1 Verification Report	Vectors, timer source and level choice, interrupt entry/exit, measured results, and a second writer to the comparator that stalled the tick for 183 ms
UM-NATOS-009	Milestone 2 Verification Report	Task model, context frame, scheduler, the zero-overhead <code>LOOP</code> defect, watchdog correction, and ageing to bound the starvation strict priority allows
UM-NATOS-010	Milestone 3 Verification Report	Heap allocator, arena model and bounds checking, and the measured DRAM budget
UM-NATOS-011	Flash Cache and Read-Only Data Placement	Mapping <code>.rodata</code> into flash, the 0x20 page congruence, and the cache-off hazard
UM-NATOS-012	Milestone 4 Verification Report	Register-based bytecode VM, the ISA, the assembler, containment of malformed programs, and the syscalls added since

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UM-NATOS-013	Milestone 5 Verification Report	Application table and lifecycle, three-level scheduling, the shell, the escape attempt, and messaging without shared memory
UM-NATOS-014	Locking Primitives	Critical sections vs blocking mutex, task blocking and idle, a measured starvation defect, and why contention cost is the number of blocking events rather than the time held
UM-NATOS-015	Display Driver	ILI9341, span rendering with no framebuffer, and the path from bit-banged SPI to SPI2 with DMA
UM-NATOS-016	Display Syscalls, and a Total System Freeze	Per-application viewports, pointer discipline, measured containment, and a yield that stopped the clock
UM-NATOS-017	Touchscreen, and a Verification Method That Failed Three Times	XPT2046 over PENIRQ, the GPIO two-bank bug, calibration by controlled input, why the four corners were the wrong four points, a capture that erased its own evidence, input confinement, and an axis that was inverted for three months because the direction test read its own worst sample
UM-NATOS-018	Persistence, and a Read Defect That Looked Like the Wrong Thing	SPI flash driver, a checksummed record that survives a power cycle, the inherited clock divider that shifted every read by a bit, and two hypotheses tested against stale firmware
UM-NATOS-019	Failure Handling, and Three Mechanisms That Had Never Fired	Stack guards enforced in the scheduler, the watchdog erasing its own panic reports, measured stack margins, a UART receive path one byte behind, and a fault reported to flash and to the panel
UM-NATOS-020	microSD over SPI, and a Pad Table That Is Not in Pin Order	SPI mode and why, per-stage error codes, the IO_MUX entry that is the UART's receive pad, and a cross-check whose samples could not have caught it
UM-NATOS-021	The Launcher, and Four Defects It Found by Existing	Icon grid and hybrid cursor, launch by name, status text that lied about state, selection read at the worst moment, the missing idle task, a close button an application cannot reach, and a correct diagnosis fixed at twenty times the necessary scope
UM-NATOS-022	The Note Pad, and a Workaround Wearing a Costume	Multi-tap keypad, messages saved to flash and read back after a power cycle, and a keypad whose key size was a touch calibration fault in disguise, plus what that fault actually was
UM-NATOS-023	The Interrupt Matrix, and Four Ways to Deliver an Edge to Nobody	The first peripheral interrupt this kernel routes: sources vs lines, a shared vector that could only have been right by accident, a PRO CPU enable bit that had to be measured, and four failures in which an edge was delivered perfectly to something that could not act on it
UM-NATOS-024	The ADC, and a Wrong Bit in a Right Register	SAR ADC1, the board's light sensor confirmed rather than assumed, and a one-bit error invisible to every read-back because it sat inside a twelve-bit field
UM-NATOS-	I2C, and Why a Preempted Master Is	Bit-banged two-wire master on the last two free pins, clock stretching as the reason preemption is safe, and a self-test for the failure where a missing pull-

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025	Legal	up makes every address answer
UM- NATOS- 026	The Shell on the Panel, and Not a Menu of It	The real shell reached from a multi-tap keypad rather than a second command set that would drift from it; output captured by teeing the UART for exactly one command

Reading order

New to the project: **001** → **002** → **004** → **003**. That gives the shape of the system, how it starts, where things live, and why the calling convention is unusual.

Picking up implementation work: **006** (what is known-good) then **007** (what is next and what it will break).

Reproducing a build: **005** alone is sufficient.

Status summary as of this revision

Item	State
Milestone 0 — kernel boots, self-checks pass	Complete, verified on hardware
Milestone 1 — timer interrupt, tick counter	Complete, verified on hardware
Milestone 2 — native task switching	Complete, verified on hardware — 3,400+ switches, zero corruption
Milestone 3 — heap and arena model	Complete, verified on hardware — all three exit criteria
Milestone 4 — bytecode interpreter	Complete, verified on hardware — program runs, six fault classes contained
Isolation	Live — every VM memory access bounds-checked, UM-NATOS-012 §6.2
Full stack	Running — bytecode hosted in a preemptible native task, 3.3M instructions across 450 preemptions with zero accounting drift, UM-NATOS-012 §6.6
Milestone 5 — multiple applications	Complete, verified on hardware — all three exit criteria
Roadmap M0–M5	Complete. Six native tasks, three scheduling levels, a shell, and an application deliberately written to escape its arena that could not
Locking	Complete — critical sections and a blocking mutex; heap and console both arbitrated, UM-NATOS-014
Task blocking	Live — <code>TASK_BLOCKED</code> , <code>TASK_SLEEPING</code> and an idle task using <code>WAITI</code> , UM-NATOS-009 §11
Scheduling	Priorities with ageing — three levels plus ageing credit, so a ready task reaches the front within ~600 ms whatever sits above it; <code>task_sleep()</code> and priority inheritance, UM-NATOS-009 §11.4
3D renderer	Running — grid raycaster, one ray per column, face shading, framebuffer on (§5.7's contrary measurement was taken through a stalling clock and lock contention), UM-NATOS-015 §5.8
Display	Working on hardware — ILI9341, no framebuffer, 43 ms full-screen fill via SPI2 + DMA, UM-NATOS-015 §5.5
Application graphics	Live — <code>FILL / TEXT / DIMS</code> confined to per-application viewports; 0 escapes across 136 audited fills, UM-NATOS-016 §5
Touch	Working on hardware — XPT2046 gated on PENIRQ and pressure; X axis was inverted for three months; calibrated on-device from four inset targets and persisted across reset, UM-NATOS-017 §4.1, §7.4

Item	State
Interrupts	Matrix verified, no consumer — peripheral interrupts routable for the first time; PENIRQ proven by injection but never observed to fire from a finger, so touch stays polled, UM-NATOS-023 §6, §7
ADC	Working on hardware — SAR ADC1, 8 channels; the light sensor on GPIO34 moved 265 counts against a 47-count control group, UM-NATOS-024 §6
I2C	Bus verified, nothing attached — bit-banged on GPIO22/27; both lines drive and release correctly, no byte has yet been transferred, UM-NATOS-025 §7, §8
On-screen shell	Working on hardware — all 25 commands reachable with no host attached; one command set, not two, UM-NATOS-026 §2
Application input	Live — <code>SYS TOUCH</code> confined to the asking application's viewport; 81 delivered, 109,211 withheld, 0 confinement failures, UM-NATOS-017 §8
Application messaging	Live — copied through a kernel mailbox, never shared memory; 278 sent / 277 delivered / 0 bad buffers, UM-NATOS-013 §8
Application bitmaps	Live — <code>SYS BLIT</code> , arena-bounded source and viewport-clipped destination, UM-NATOS-012 §10
DRAM budget	Measured at M3 — 167,680 B allocatable then, 158,000 B after <code>TASK_MAX</code> rose to 8; it has since risen to 12 and three drivers have been added, so the current figure is lower and unmeasured, UM-NATOS-010 §7.2
Flash cache	Enabled — <code>.rodata</code> mapped from flash, UM-NATOS-011
Persistence	Live — checksummed record in a flash sector at 2 MB; boot counter and a cumulative frame count survived 16 resets, UM-NATOS-018 §6
Failure handling	Enforced — guards checked on every switch across every task slot; <code>hang</code> / <code>fault</code> / <code>smash</code> each trigger their path on demand, UM-NATOS-019 §5
Fault reporting	Three ways — flash record read back by the next boot, UART report, and the reason drawn on the panel; ordered by decreasing reliability, UM-NATOS-019 §6–7
microSD	Reading — SPI mode, per-stage errors, bounded on an empty slot; FAT16 header read at LBA 240, UM-NATOS-020 §5.2
Launcher	Live — 3×3 icon grid including an on-screen shell, hybrid cursor, double-tap to launch by name; close button per program in a column outside every viewport, so a program cannot hide its own exit, UM-NATOS-021 §6.2
Notes	Live — multi-tap keypad, 8 messages of 160 characters in flash, verified across a power cycle and a reflash, UM-NATOS-022 §6
Stack margins	Measured — worst task uses 444 B of 2,048; minimum margin 78%, UM-NATOS-019 §3.1
Version control	Initialised 2026-08-14
JTAG debug probe	Ordered, not in hand

Item	State
Bootloader IRAM overlap	Closed — investigated and disproved, UM-NATOS-004 §5
Panic handler	Closed — exercised deliberately with an <code>111</code> instruction; prints EXCCAUSE/EPC and halts
Watchdogs	Closed — measured armed, now disabled and read back, UM-NATOS-009 §8

Standing rule for interrupt handlers — clear `PS.EXCM` before calling C. Hardware sets `EXCM` on interrupt entry, and while it is set the Xtensa zero-overhead loop-back is disabled: a hardware loop body executes once and falls through to whatever sits at `LEND`. GCC emits these loops for ordinary counted C loops, so this silently degrades **every C function reachable from a handler** — drivers and the VM interpreter included, not just the scheduler where it was found. `_handler_level3` now clears it; any future handler must do the same. UM-NATOS-009 §6.

Standing rule for shared registers — a hardware register with a software shadow has one safe shape: either one writer, or every writer maintains the shadow. `timer.c` kept `g_next` as its idea of the comparator deadline while `task_yield()` wrote `CCOMPARE1` directly. The handler then added a whole interval to a deadline that was only 64 cycles old, so every yield pushed the tick further out — 18 tick periods, 183 ms, before anything noticed. The rule below was obeyed exactly and did not prevent it, because it constrains the WRITER and the defect was in the other party's bookkeeping. UM-NATOS-008 §8.

Standing rule for the scheduler — a yield must never defer the clock it depends on. `task_yield()` originally wrote `CCOMPARE1 = ccount + 64` unconditionally, so any loop yielding faster than 64 cycles pushed the deadline ahead of `CCOUNT` forever, the timer interrupt stopped firing, and the whole kernel froze — the tick is what drives every context switch. Any routine adjusting a scheduler deadline must only ever move it **earlier**. UM-NATOS-016 §3.

Standing rule for verification — when an instrument reports its own reading invalid, that outranks every value printed beside it. Three times across two sessions a frozen marker, a sample counter stuck at an identical value, and a boot banner in a serial capture each said "this measurement is not what you think", and the plausible-looking numbers next to them were believed anyway. Two conclusions were published and later retracted as a result. Latch the quantity so timing cannot lie about it, then feed the system a controlled input rather than interpreting an uncontrolled one. UM-NATOS-017 §6.

Standing rule for debugging — a negative result is only informative if the experiment demonstrably ran. Two flash hypotheses were recorded as tested against a board that had never been reflashed: `build.ps1` builds, but `build.ps1 -Flash` is what flashes, and the invocation used named a script that does not exist. Every hypothesis returned bit-identical output, which was read as "none of these are the cause" when it meant "no experiment has run yet". The signature to watch for is **a run of results that do not vary when the input does** — suspect the harness before the theory, and verify the change reached the target rather than inferring it from the absence of an error. UM-NATOS-018 §5.

Standing rule for verification — a startup artefact is not evidence the thing it introduces works. The shell was signed off because its banner printed; the banner proves a task was created and the TRANSMIT path works, and says nothing about receive. The receive path was one byte behind for the shell's entire existence, so pressing Enter did nothing until the next keystroke — and every automated test drove it with CR and LF, the one input shape that hides it. Stack guards and the panic handler went unexercised for the same reason: each was confirmed to EXIST rather than observed to WORK. Trigger the mechanism on purpose, or treat it as untested. UM-NATOS-019 §4.1.

Standing rule for calibration — never infer direction from the endpoint of a gesture. The first and last samples of a drag are its two least trustworthy, because both sit at a contact transition where the panel is not bridged and the ADC reads its rail. A rail reading is near the top of the range, so a drag ending anywhere "ends near its maximum" and EVERY axis appears to increase — the test can only return one answer. The touch X axis was backwards for three months behind exactly that. Calibrate from labelled points, each a known position paired with a reading, and let direction fall out of comparing labels. UM-NATOS-017 §7.1.

Standing rule for lock contention — the cost is the NUMBER of blocking events, not the time the lock is held. Each one costs a scheduling round-trip whether the lock was held for 24 ms or 24 µs: the blocked task is descheduled and must be selected again. Measured here at 63 ms per contention against a 24 ms hold, with the lock free 88% of the time while two tasks sat blocked on it. Narrowing the hold by 25% changed the outcome by nothing. The levers are batching (fewer takes) or not blocking at all (`try_lock`) — shortening holds, the intuitive move, does nothing. UM-NATOS-014 §10.5.

Standing rule for cross-checks — a check only tests what its samples can distinguish. The SD pin map was verified against four IO_MUX entries already in the tree. All four confirmed the indexing, and all four sat below the anomaly, so every one of them agreed with the wrong answer: GPIO23 was read from the UART's receive pad. The check was real, it was performed, and it could not have failed. Samples must STRADDLE the thing being verified, not merely agree with it. UM-NATOS-020 §4.2.

Standing rule for fixes — a correct diagnosis does not license a fix of arbitrary scope. Chrome drawn over a view that repaints every pixel every frame will strobe; that diagnosis was reached twice and was right both times. The first fix reserved rows for it, which moved the region boundary, the application strips, the colour strip and the grid — four files of constants — and produced a screen that looked wrong for reasons never found, while every measurement insisted the renderer was correct. The second wrote 324 pixels into a buffer that was already being sent. Prefer the fix whose blast radius matches the defect. UM-NATOS-021 §6.6.

Standing rule for workarounds — tolerating a defect is not fixing it. The note pad's keys are 80 px because the touch mapping reads about 24 px low on X; a 24 px key was destroyed by that error and an 80 px key absorbs it. The app works, the fault is untouched, and every future element finer than 80 px will meet it again. Record which one you did, at the place a reader would otherwise assume the generous version. UM-NATOS-022 §3.3.

Standing rule for reverts — reverting two changes together destroys the information about which one mattered. A relay layout and a camera fix were reverted in one commit; the screen looked right afterwards, which appeared to convict the layout. Re-applying the geometry alone, later, showed it was innocent — the camera had broken concurrently and "blank screen" was a correct rendering of the inside of a wall. If a revert must bundle, the bundle is a hypothesis to test later, not a conclusion. UM-NATOS-021 §6.7.